

Talha Khan

Assistant Professor

College of Computing, Grand Valley State University, USA

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<https://talhakh13.github.io/> [Google Scholar](#)

RESEARCH OVERVIEW

My work lies in the general area of **human-computer interaction** (HCI). The goal of my research is to develop **mixed reality** (MR) systems that solve real-world problems by seamlessly integrating information into the physical world through virtual interfaces. I then study how these systems impact users across a range of tasks, including surgical procedures, sensemaking, and everyday activities. My work has been published in leading venues such as ACM CHI, ACM ISS, ACM SUI, and IEEE ISMAR, where it received a Best Paper Award (2023).

ACADEMIC EXPERIENCE

- **Assistant Professor** AUG. 2025 - CURRENT
College of Computing, Grand Valley State University, USA
• Director, Immersive Interaction Lab
- **Graduate Research Fellow** SEP. 2024 - APR. 2025
Department of Computer Science, University of Pittsburgh, USA
- **Research Intern, Human-Centered AI** MAY. 2024 - AUG. 2024
Toyota Research Institute, USA
- **Research Intern, Augmented Perception Lab** MAY. 2023 - AUG. 2023
Carnegie Mellon University, USA
Mentor: [DR. DAVID LINDLBAUER](#)
- **Graduate Student Researcher** MAY. 2021 - APR. 2024
Department of Computer Science, University of Pittsburgh, USA

EDUCATION

- **PhD in COMPUTER SCIENCE** AUG. 2020 - APR. 2025
University of Pittsburgh, USA
• Thesis: Towards Integrating Mixed Reality Into Physical Workspaces
• Advisor: [DR. JACOB BIEHL](#)
- **MS in COMPUTER SCIENCE** SEP. 2018 - JUN. 2020
University of California, Irvine, USA
- **BS in COMPUTER SCIENCE** AUG. 2014 - JUN. 2018
Lahore University of Management Sciences, Pakistan

TEACHING EXPERIENCE

- **Course Instructor** AUG. 2025 - CURRENT
Grand Valley State University, USA
Courses: *Computer Science 1* (Introduction to Python)
- **Graduate Teaching Assistant** SEP. 2020 - APR. 2021
University of Pittsburgh, USA
Courses: *Algorithm Implementation, Programming Language for Web Applications*

- **Graduate Teaching Assistant** OCT. 2018 - MAY. 2020
University of California, Irvine, USA
Courses: *Intermediate Programming (Python), Data Structures (C++), Introduction to C++*
- **Undergraduate Teaching Assistant** SEP. 2017 - MAY. 2018
Lahore University of Management Sciences, Pakistan
Courses: *Applied Probability (Graduate-level), Probability*

PUBLICATIONS

Conference Publications & Proceedings

- [C.10] ReadAR: Utilizing Mixed Reality for Physical Document Digitization
Liam Adams, **Talha Khan**
ACM IMX 2026
- [C.9] Utilizing VLMs for Identifying Child Safety Hazards
Talha Khan, Ahmed Ashraff Butt
ACM CHI 2026
- [C.8] MR-MDEs: Exploring the Integration of Mixed Reality into Multi-Display Environments
Griffin Hurt, **Talha Khan**, Nicolás Kass, Anthony Tang, Edward Andrews, Jacob Biehl
PACM HCI (ACM ISS 2025)
- [C.7] Shadows of Reality: Enhancing Bystander Awareness of Mixed Reality Interfaces
Talha Khan, Abigail Zimmerman, Edward Andrews, David Lindlbauer, Jacob Biehl
ACM SUI 2025
- [C.6] Don't Block My Stuff: Fostering Personal Object Awareness in Multi-user Mixed Reality Environments
Talha Khan, David Lindlbauer
PACM HCI (ACM ISS 2024)
- [C.5] Understanding Perceived Utility and Comfort of In-Home General-Purpose Sensing through Progressive Exposure
Pranut Jain, Andrew Xu, Thomas Downes, Injung Kim, **Talha Khan**, Jacob Biehl, Adam Lee.
PACM HCI (ACM CSCW 2024)
- [C.4] EndovascularAR: Utility of Mixed Reality to Segment Large Displays in Surgical Settings
Griffin Hurt, **Talha Khan**, Michael Kann, Edward Andrews, Jacob Biehl
IEEE VRW 2024
- [C.3] Future of Surgical Mixed Reality: Cutting-Edge or Cutting Too Deep?
Talha Khan, Edward Andrews, Jacob Biehl
IEEE VRW 2024
- [C.2] Understanding Effects of Visual Feedback Delay in AR on Fine Motor Surgical Tasks
Talha Khan, Toby Zhu, Thomas Downes, Lucille Cheng, Nicolás Kass, Edward Andrews, Jacob Biehl.
IEEE TVCG (ISMAR 2023) Best Paper Award
- [C.1] AR in the OR: exploring use of augmented reality to support endoscopic surgery
Talha Khan, Edward Andrews, Paul Gardner, Arka Mallela, Jeffrey Head, Joseph Maroon, Georgios Zenonos, Dmitriy Babichenko, Jacob Biehl
ACM IMX 2022

Journal Publications

- [J.5] Application of Mixed Reality in Endoscopic Streaming during Skull Base Surgery
Yujin Choi, Shovan Bhatia, Adam Monek, Harrison Alexander, **Talha Khan**, Brendan Szczepkowski, Griffin Hurt, Anthony Tang, Michael Kann, Regan Shanahan, Brock Gjesdal, Rishi Basdeo, Kyle Affolter, David Fernandes Cabral, Georgios Zenonos, Paul Gardner, Jacob Biehl, Edward Andrews
Journal of Neurological Surgery Part B: Skull Base 2026 (Abstract)
Operative Neurosurgery 2026 (Full Paper)
- [J.4] Evolution of the meta-neurosurgeon: A systematic review of the current technical capabilities, limitations, and applications of augmented reality in neurosurgery
Nikhil Sharma, Arka Mallela, **Talha Khan**, Stephen Paul Canton, Nicolás Kass, Fritz Steuer, Jacquelyn Jardini, Jacob Biehl, Edward Andrews.
Surgical Neurology International 2024

- [J.3] Solar-powered Parking Analytics System using Deep Reinforcement Learning
Yoones Rezai, **Talha Khan**, Stephen Lee, Daniel Mossé
ACM TOSN 2023
- [J.2] A systematic comparison of the accuracy of monocular RGB tracking and LiDAR for neuronavigation
Talha Khan, Jacob Biehl, Edward Andrews, Dmitriy Babichenko
Healthcare Technology Letters 2022
- [J.1] Streaming 2D-Endoscopic Video into an Augmented Reality Headset Display: A Feasibility Study
Edward Andrews, **Talha Khan**, Arka Mallela, Joseph Maroon, Jacob Biehl, David Fernandes-Cabral, Zachary Gersey, Hussam Abou-Al-Shaar, Paul Gardner, Georgios Zenonos
Journal of Neurological Surgery Part B: Skull Base 2022

MENTORSHIP EXPERIENCE

- Khadia Arif JAN. 2026 - CURRENT
MS Human-Centered Computing, Grand Valley State University, USA
Project: *ProjectAR: Externalizing Mixed Reality Interfaces to Support Bystander Collaboration*
- Liam Adams AUG. 2025 - CURRENT
BS Integrative Studies, Grand Valley State University, USA
Project: *ReadAR: Utilizing Mixed Reality for Physical Document Digitization*
- Abigail Zimmerman SEP. 2024 - JAN. 2025
BS Digital Narrative and Interactive Design, University of Pittsburgh, USA
Project: *Shadows of Reality: Enhancing Bystander Awareness of Mixed Reality Interfaces (ACM SUI 2025)*
- Bo-Chen Kuo JAN. 2024 - APR. 2024
MS Computer Science, University of Pittsburgh, USA
Project: *Automatic Segmentation of Medical Appliance Buttons*
- Elliott Finney JAN. 2024 - APR. 2024
BS Computer Science, University of Pittsburgh, USA
Project: *Multi-modal Mixed Reality Interfaces for Medical Appliances*
- Griffin Hurt MAY. 2023 - APR. 2024
BS Computer Science, University of Pittsburgh, USA
Project: *MR-MDEs: Exploring the Integration of Mixed Reality into Multi-Display Environments (ACM ISS 2025)*
- Chloe Dahan JUN. 2021 - AUG. 2021
BS Digital Narrative and Interactive Design, University of Pittsburgh
Project: *Automatic Human-Humanoid Gesture Imitation*
- Aiden Dorneich JUN. 2021 - AUG. 2021
Research Intern Facet Lab, University of Pittsburgh, USA
Project: *Automatic Human-Humanoid Gesture Imitation*

HONORS AND AWARDS

- CS 50 Fellowship Award, Department of Computer Science, University of Pittsburgh APR. 2024
- Best Journal Paper Award, IEEE International Symposium on Mixed and Augmented Reality OCT. 2023
- Honorable Mention Award, Link Foundation Modelling, Simulation & Training Fellowship JUN. 2022

MEDIA APPEARANCES

- **Pitt To The Point** MAR. 2024
[*Revolutionizing Healthcare With Mixed Reality*](#)

INVITED TALKS

- **Dean's Spotlight Series, University of Pittsburgh** JAN. 2023
AR in the OR: An Interdisciplinary Journey Towards the Holographic Operation Room

LEADERSHIP EXPERIENCE

- **Business Manager** MAR. 2024 - APR. 2025
Doctoral Guild at Pitt, University of Pittsburgh, USA
- **Founding Member** APR. 2023 - CURRENT
Surreality Lab, University of Pittsburgh, USA

REVIEWING SERVICE

- ACM International Conference on Interactive Surfaces and Spaces (ACM ISS 2026)
- Springer Nature Virtual Reality (2026)
- IEEE International Conference on Virtual Reality and 3D User Interfaces (*IEEE VR 2026*)
- ACM CHI Conference on Human Factors in Computing Systems (*ACM CHI 2026*)
- ACM Symposium on Spatial User Interaction (*ACM SUI 2025*)
- ACM CHI Conference on Human Factors in Computing Systems (*ACM CHI 2025*)
- ACM Symposium on User Interface Software and Technology (*ACM UIST 2024*)
- ACM CHI Conference on Human Factors in Computing Systems (*ACM CHI 2024*)
- ACM CHI Conference on Human Factors in Computing Systems (*ACM CHI 2023*)
- ACM Conference On Computer-Supported Cooperative Work And Social Computing (*ACM CSCW 2023*)

VOLUNTEER SERVICE

- Student Volunteer, ACM Conference on Interactive Surfaces and Spaces (ACM ISS 2023)
- Student Volunteer, ACM Conference on Human Factors in Computing Systems (ACM CHI 2022)

REFERENCES

DR. JACOB BIEHL

Associate Professor, Department of Computer Science, University of Pittsburgh, USA

🌐 <https://jtbiehl.github.io/> | ✉ biehl@pitt.edu

DR. DAVID LINDLBAUER

Assistant Professor, Human-Computer Interaction Institute, Carnegie Mellon University, USA

🌐 <https://www.davidlindlbauer.com/> | ✉ davidlindlbauer@cmu.edu

DR. ERIN WALKER

Professor, Department of Computer Science, University of Pittsburgh, USA

🌐 <https://erinwalker.academicwebsite.com/> | ✉ eawalker@pitt.edu

DR. EDWARD ANDREWS

Assistant Professor, Department of Neurological Surgery, University of Pittsburgh, USA

✉ andrewse2@upmc.edu